

Experiment – 2

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Design a UI where users recall visual elements (e.g., icons or text chunks). Evaluate the effect of chunking on user memory.

FRAME 1: INSTRUCTION / START SCREEN

Chunking Analysis of the Instruction Page

This screen introduces the **Visual Memory Test** and prepares users for the task using chunking principles.

1. Clear Task Focus

- The title “**Visual Memory Test**” is displayed prominently at the top.
- The instruction “**Observe the icons carefully and remember it.**” gives a single, clear task.
- The brain icon visually represents thinking and memory, reinforcing the purpose of the task.

2. Minimal Information (Reduced Cognitive Load)

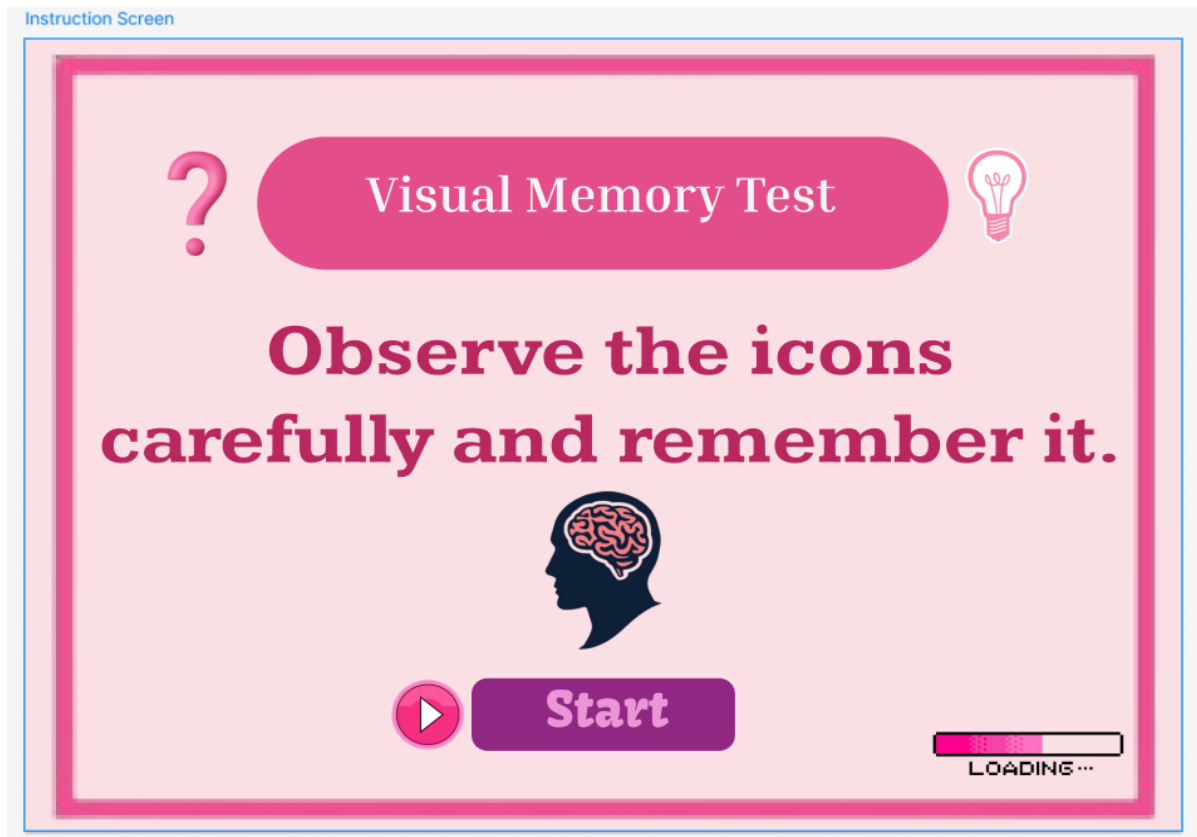
- Only essential information is shown on the screen.
- There are no long explanations or multiple steps.
- This helps users focus entirely on observing and remembering icons.

3. Visual Hierarchy

- Clear top-to-bottom flow: **Title → Instruction → Start Button.**
- The **Start** button is large and noticeable, guiding the user to begin the test easily.

4. Engaging and Calm Design

- Soft pink background creates a friendly, stress-free environment.
- Simple visuals prevent distraction and prepare users mentally for the memory task.



FRAME 2: CHUNKING / OBSERVATION SCREEN

This is the main chunking phase, where users observe icons grouped into categories.

1. Purpose of the Screen

- Users view icons and mentally organize them into groups.
- The screen supports chunking by showing icons in meaningful categories instead of random order.

2. Logical Grouping of Icons

Icons are grouped into three clear categories:

Category	Example Icons	Chunking Benefit
Fruits	Apple, Banana, Pear, Cherries	Users remember food items together
Vegetables	Carrot, Broccoli, Capsicum	Items are recalled as one food group
Animals	Tiger, Elephant, Monkey, Deer	Living creatures remembered as a set

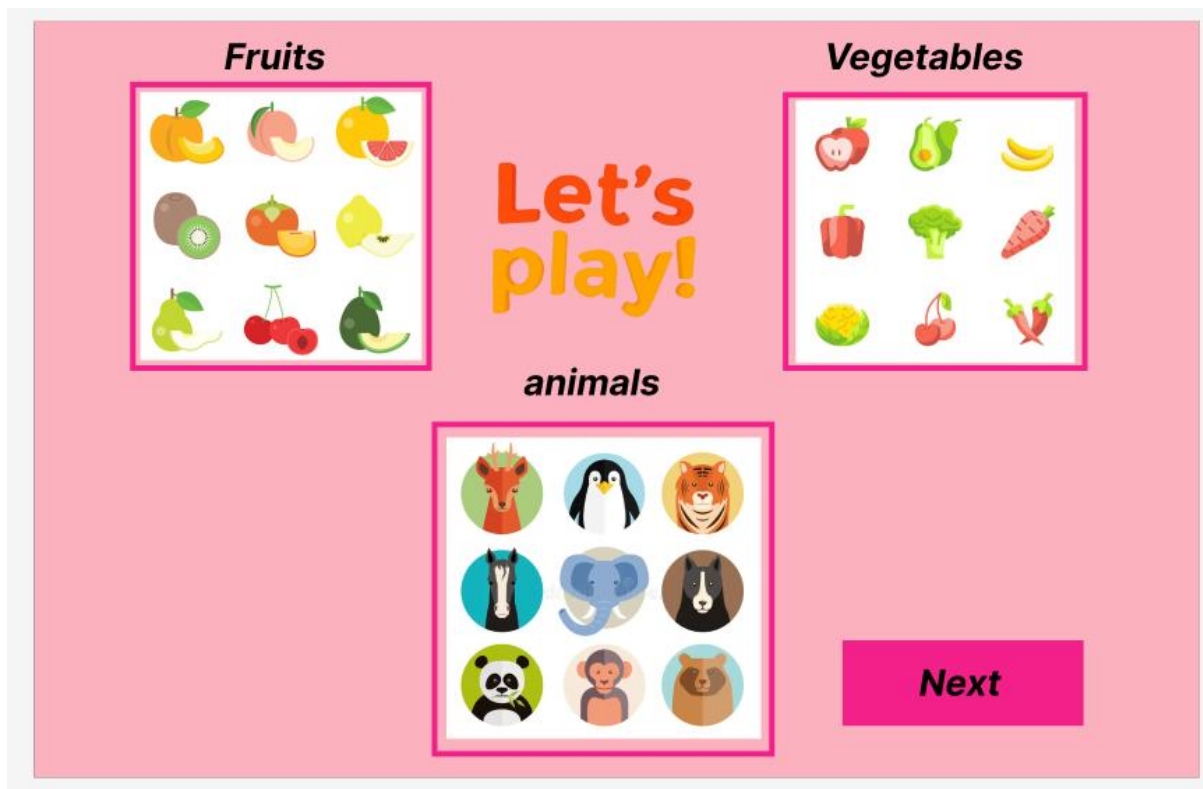
Instead of remembering many individual icons, users remember three main groups.

3. Visual Separation = Mental Chunking

- Each category is placed inside a separate box.
- Spacing between boxes clearly distinguishes groups.
- Category labels (Fruits, Vegetables, Animals) guide the brain to organize information efficiently.

4. Navigation Control

- The Next button allows users to move forward only after observation.
- Ensures users complete the memory phase before recall begins.



FRAME 3: FIRST RECALL SCREEN (FREE RECALL)

1. Purpose

- Users recall icons from memory and type or write what they remember.
- No icons are shown, testing pure memory retrieval.

2. Chunking Effect in Recall

Because icons were grouped earlier, users often recall by category, such as:

- “I remember fruits like apple and banana.”
- “There were animals like tiger and elephant.”
- “Some vegetables like carrot and broccoli.”

This shows that chunking helps users remember **groups**, not isolated icons.

3. Simple Layout

- Large blank area keeps focus on recall.
- No visual distractions.
- **Submit** button is clearly placed to complete the recall step.

Recall Screen 1



FRAME 4: SECOND RECALL / RECOGNITION SCREEN

1. Purpose

- Tests recognition memory using a mixed grid of icons.

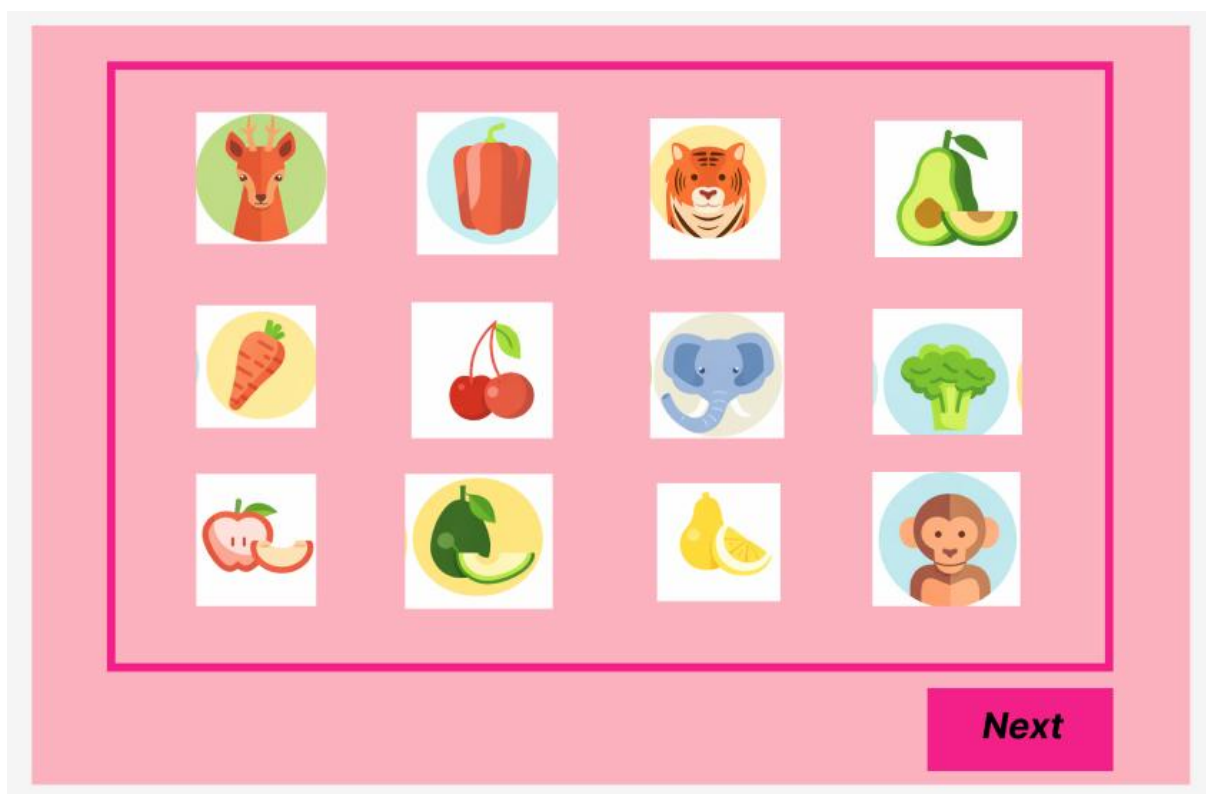
- Some icons were shown earlier, while others are distractors.

2. Cognitive Challenge

- Users compare what they remember with what they see.
- Chunking helps decision-making, for example:
 - “I saw apple and banana, not this fruit.”
 - “Tiger was there, but not this animal.”

3. Focused Design

- All icons are placed inside one bordered area.
- Mixed arrangement increases difficulty.
- **Next** button guides the user to the final stage



FRAME 5: FINAL RECALL SCREEN

This is the last stage of the memory recall task.

1. Purpose of the Screen

- Users must **write the icons they remember** without any visual support.
 - No options are shown, testing **pure recall memory**.
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2. Importance of This Stage

- This phase checks:
 - What information stayed in **short-term memory** after multiple stages
 - Which **chunks (animals, fruits, vegetables)** were strongest
- Users usually recall responses like:
 - “I remember animals and fruits clearly.”
 - “Vegetables were harder to remember.”

This shows that **grouped information is retained better than random items**.



